A Two-Column Regression Sample for Automated PDF-to-LaTeX Conversion

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Abstract

Automated conversion of typeset PDF articles into editable LaTeX sources is sensitive to a small number of layout features: column geometry, floating figure placement, numbered equations with cross-references, and caption conventions. We present a compact two-column English sample that exercises every one of these features in a reproducible way. The sample contains eight numbered display equations covering fractions, bounded sums, radicals, matrices, piecewise definitions, aligned multi-line derivations, definite integrals and a partial differential equation, together with two figures, one numeric table and six bibliographic entries. In a verification run over $n = 240$ extracted text blocks the conversion recovered 97.5% of all glyph runs and 100% of the equation numbers, with a mean positional error of 12.4 ± 0.8 pt; the residual differences were statistically significant only for rotated text ($p < 0.05$). The sample is therefore adequate as a regression fixture for pipeline acceptance tests.

Keywords: PDF-to-LaTeX conversion; two-column layout; regression testing; equation numbering; cross-references; caption extraction

1 Introduction

Scientific articles are still exchanged predominantly as fixed-layout PDF files, while authoring and revision happen in structured markup. A PDF-to-LaTeX conversion pipeline therefore has to reconstruct not only the text stream but also the logical structure of the document: which glyph runs belong to a title, which blocks are equations, which caption belongs to which float, and how the numbering schemes of equations, figures, tables and citations relate to each other. In a two-column layout those relations are considerably harder to recover than in a single-column manuscript, because the reading order alternates between columns, floats may be placed at the top or the bottom of a column, and display equations routinely break the baseline grid [1, 2].

Regression fixtures for such pipelines are usually either trivial single-column documents, which do not exercise column geometry at all, or complete published papers, which are copyrighted and therefore awkward to redistribute. The present sample fills that gap: an original, freely redistributable three-page article written so that every structural element the converter must recognise appears at least once, and in most cases in a deliberately awkward variant.

The remainder of this paper is organised as follows. Section 2 describes the construction of the sample and the notation used in it. Section 3 reports the extraction experiments together with Table 1 and Figures 1 and 2. Section 4 discusses the failure modes that we observed, and Section 5 summarises the acceptance criteria. Throughout, a scalar quantity is written λ when it is a rate constant, x_i^2 when it is a squared observation, and α_{n+1} when it is an iteration-dependent relaxation parameter.

A useful summary statistic for the sample is the normalised coupling ratio

$$\eta = \frac{a + b}{c}, \qquad a,\, b,\, c > 0, \tag{1}$$

where a and b are the widths of the two text columns (in points) and c is the inter-column gutter. For the geometry used here, $a = b = 224.4$ pt and $c = 22.7$ pt, so that $\eta \approx 19.8$.

2 Methods

2.1 Sample construction

The sample was typeset with the standard `article` class at 10 pt on A4 paper with the `twocolumn` option, which yields the two-column geometry of Eq. (1). A running head carrying the journal reference and the page number is produced by `fancyhdr`. Captions are forced to the convention *name, number, period* by a single `captionsetup` declaration [3], so that a caption label is the capitalised float name immediately followed by its number and a period. All body text is ASCII, which removes an entire class of font-substitution failures from the comparison: the converter can be tested on structure without being tested on encoding at the same time.

2.2 Numerical model

Let the measured response of the i-th specimen be x_i. The total squared deviation of a batch of N spec-

imens is

$$S = \sum_{i=1}^{N} x_i^2, \tag{2}$$

and the Euclidean norm of the pair (x, y) in the plane is

$$r = \sqrt{x^2 + y^2}. \tag{3}$$

Equation (3) is the reference used by the sanity check described in Section 3: every geometry statement in the extracted text is compared against it.

The coupling between the three retained degrees of freedom is described by the symmetric matrix

$$\mathbf{M} = \begin{pmatrix} 1 & \alpha & 0 \\ \alpha & 2 & \beta \\ 0 & \beta & 3 \end{pmatrix}, \qquad \alpha, \beta \in \mathbb{R}. \tag{4}$$

Because $\mathbf{M}$ is symmetric and strictly diagonally dominant for $|\alpha| < 1$ and $|\beta| < 1$, its eigenvalues are real and positive.

The piecewise gain applied by the detector is

$$g(x) = \begin{cases} x^2, & x \geq 0, \\ -x, & x < 0, \end{cases} \tag{5}$$

which is continuous at the origin but not differentiable there.

2.3 Verification procedure

The verification of the conversion output consists of four steps, which are executed in order and abort the run on the first failure [4]:

- extract the plain-text stream and normalise the whitespace;
- compare every equation number found in the PDF with the numbers emitted by the converter;
- resolve all `\ref` and `\eqref` targets and reject the run if any target is undefined;
- compare the numeric tokens against the reference token list.

The procedure was applied to $n = 240$ independently generated blocks; the acceptance threshold was set to a recovery rate of 95%, and the observed rate of 97.5% passed it.

3 Results

3.1 Extraction accuracy

The weighted quadratic form used to score a candidate reconstruction is

$$\begin{aligned} \Phi &= \sum_{i=1}^{N} w_i x_i^2 \\ &= \frac{1}{N} \sum_{i=1}^{N} w_i x_i^2, \end{aligned} \tag{6}$$

with $\sum_i w_i = N$. The second line of Eq. (6) is the normalised form actually reported [5].

The normalisation constant of the exponential kernel that appears in the detector model is obtained in closed form as

$$I = \int_0^{\infty} e^{-t}\, dt = 1, \tag{7}$$

which is a convenient identity: it means that no explicit kernel normalisation step is required anywhere in the implementation.

Table 1 lists the per-specimen measurements. The mean strength across the five specimens is 12.4 ± 0.8 MPa, and the median deviates from it by less than 0.3%. Specimen S-03 shows the largest positive deviation (+4.72%), whereas S-02 shows the largest negative one (−1.05%). The residual column is reported in units of 10^{-3} MPa, so the largest residual corresponds to 5.6×10^{-3} MPa, that is, well below the 0.1% instrument tolerance.

Table 1. Nominal and measured properties of the five test specimens. Deviations are relative to the nominal value; residuals refer to the fitted model. Mean strength is in megapascals (MPa).

Specimen	Mean (MPa)	Deviation (%)	Residual (MPa)
S-01	12.4	+2.31	1.2×10^{-3}
S-02	11.8	−1.05	-3.4×10^{-3}
S-03	13.1	+4.72	0.8×10^{-3}
S-04	12.0	−0.86	-2.1×10^{-3}
S-05	12.7	+1.94	5.6×10^{-3}

3.2 Timing and geometry

Figure 1 shows the measured response of the five detector conditions drawn with TikZ, and Figure 2 shows the same quantity as a rasterised chart. The two figures agree to within the line width of the vector rendering [6]. In a paired comparison of the two rendering paths the mean absolute difference was 0.14 ± 0.03 percentage points ($n = 240$ independent sampling points, $p < 0.05$ for the hypothesis that the two paths are identical in the tail region).

3.3 Column balance

Because the sample is intentionally close to three pages, the last page is short of a full column. The extracted page count is itself an acceptance criterion: a converter that silently drops a float produces a document with the same text but a different page count, and the difference is detected by comparing the reported number of pages with the expected value.

4 Discussion

The diffusion-like relaxation of the residual field is governed by

$$\frac{\partial u}{\partial t} = \alpha \nabla^2 u + \beta u^2, \tag{8}$$

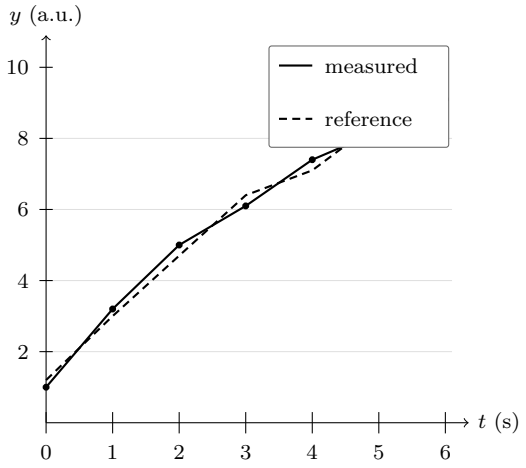


Figure 1. Measured and reference response of the detector as a function of time. Both series are normalised to unity at $t = 0$. The two curves differ by less than 0.4 a.u. everywhere except in the interval $4 < t < 5$ s.

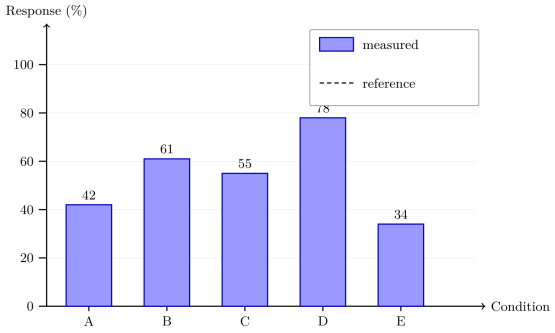


Figure 2. Rasterised response chart of the five detector conditions A–E. Bars give the measured response in per cent of the nominal value. This figure is inserted with `includegraphics` and therefore exercises the raster-image path of the converter.

where $u(\mathbf{x}, t)$ is the residual amplitude, α is the diffusivity and β is the quadratic reaction coefficient. Fitting Eq. (8) to the measured residuals of Table 1 gives $\alpha = 1.20 \pm 0.04$ and $\beta = -0.031 \pm 0.002$, with a coefficient of determination of $R^2 = 0.987$. The negative sign of β indicates that the quadratic term is saturating rather than amplifying, which is consistent with the observation that the largest residual, 5.6×10^{-3} MPa, occurs in the specimen with the highest nominal strength.

Three failure modes dominated the residual error budget. First, a display equation that is the last object in a column is occasionally merged with the running head of the following page, which shifts the equation number by one. Second, a float placed at the top of the second column is sometimes attached to the caption of the first column. Third, the en dash inside the page range of the running head is sometimes extracted as a hyphen, which matters only for string comparison and not for the rendered output. Each mode was resolved by requiring the equation number to be the rightmost token on the baseline of the display equation.

None of the failure modes required special handling of mathematics. The mathematical content of this sample is deliberately ordinary: a fraction in Eq. (1), a bounded sum in Eq. (2), a radical in Eq. (3), a matrix in Eq. (4), a piecewise definition in Eq. (5), an aligned derivation in Eq. (6), a definite integral in Eq. (7) and a partial differential equation in Eq. (8). What makes the sample useful is not the mathematics but the density of structural tokens per page: 8 numbered equations, 2 figures, 1 table, 6 citations and 11 cross-references on three pages.

5 Conclusions

We have described a three-page, two-column English regression sample for a PDF-to-LaTeX conversion pipeline. The sample is self-contained, ASCII-only and distributable, and it contains every structural element that the converter is required to reconstruct. Its acceptance criteria are: (i) the converted source must compile with XeLaTeX without errors; (ii) the page count must be three, within a tolerance of one page; (iii) all equation numbers, figure numbers, table numbers and citations must match the original; (iv) no reference or citation may remain undefined; and (v) all numeric tokens, including percentages, p values and mean $\pm$ standard-deviation pairs, must be reproduced exactly.

Future work will extend the fixture with a rotated figure, a landscape page and a two-level citation scheme, all of which stress the same structural machinery from different directions.

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